C: Dipan is an avid reader of Tolstoy

It is important to understand that whatever structure an English sentence might have is lost when it is symbolized by a sentence letter of TFL. From the point of view of TFL, a sentence letter is just a letter. It can be used to build more complex sentences, but it cannot be taken apart.

CHAPTER 5

Connectives

In the previous chapter, we considered symbolizing fairly basic English sentences with sentence letters of TFL. This leaves us wanting to deal with the English expressions 'and', 'or', 'not', and so forth. These are *connectives*—they can be used to form new sentences out of old ones. In TFL, we will make use of logical connectives to build complex sentences from atomic components. There are five logical connectives in TFL. This table summarizes them, and they are explained throughout this section.

symbol	**what it is called**	**rough meaning**
$\neg$	negation	'It is not the case that. . .'
$\wedge$	conjunction	'Both. . . and . . .'
$\vee$	disjunction	'Either. . . or . . .'
$\rightarrow$	conditional	'If . . . then . . .'
$\leftrightarrow$	biconditional	'. . . if and only if . . .'

These are not the only connectives of English of interest. Others are, e.g., 'unless', 'neither . . . nor . . .', and 'because'. We will see that the first two can be expressed by the connectives we will discuss, while the last cannot. 'Because', in contrast to the others, is not *truth functional*.

5.1 Negation

Consider how we might symbolize these sentences:

1. Mary is in Barcelona.
2. It is not the case that Mary is in Barcelona.
3. Mary is not in Barcelona.

In order to symbolize sentence 1, we will need a sentence letter. We might offer this symbolization key:

> B: Mary is in Barcelona.

Since sentence 2 is obviously related to sentence 1, we will not want to symbolize it with a completely different sentence letter. Roughly, sentence 2 means something like 'It is not the case that B'. In order to symbolize this, we need a symbol for negation. We will use '$\neg$'. Now we can symbolize sentence 2 with '$\neg B$'.

Sentence 3 also contains the word 'not', and it is obviously equivalent to sentence 2. As such, we can also symbolize it with '$\neg B$'.

> A sentence can be symbolized as $\neg\mathcal{A}$ if it can be paraphrased in English as 'It is not the case that...'.

It will help to offer a few more examples:

4. The widget can be replaced.
5. The widget is irreplaceable.
6. The widget is not irreplaceable.

Let us use the following representation key:

> R: The widget is replaceable

Sentence 4 can now be symbolized by 'R'. Moving on to sentence 5: saying the widget is irreplaceable means that it is not

the case that the widget is replaceable. So even though sentence 5 does not contain the word 'not', we will symbolize it as follows: '$\neg R$'.

Sentence 6 can be paraphrased as 'It is not the case that the widget is irreplaceable.' Which can again be paraphrased as 'It is not the case that it is not the case that the widget is replaceable'. So we might symbolize this English sentence with the TFL sentence '$\neg\neg R$'.

But some care is needed when handling negations. Consider:

7. Jane is happy.
8. Jane is unhappy.

If we let the TFL-sentence 'H' symbolize 'Jane is happy', then we can symbolize sentence 7 as 'H'. However, it would be a mistake to symbolize sentence 8 with '$\neg H$'. If Jane is unhappy, then she is not happy; but sentence 8 does not mean the same thing as 'It is not the case that Jane is happy'. Jane might be neither happy nor unhappy; she might be in a state of blank indifference. In order to symbolize sentence 8, then, we would need a new sentence letter of TFL.

5.2 Conjunction

Consider these sentences:

9. Adam is athletic.
10. Barbara is athletic.
11. Adam is athletic, and also Barbara is athletic.

We will need separate sentence letters of TFL to symbolize sentences 9 and 10; perhaps

A: Adam is athletic.
B: Barbara is athletic.

Sentence 9 can now be symbolized as 'A', and sentence 10 can be symbolized as 'B'. Sentence 11 roughly says 'A and B'. We